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convex shapes with an increasing fabricated distance of two rings [Fig. 1(b)]. Managing these varied responses requires a wavelength tracking system capable of identifying and adapting to different shapes.

The RX adaptive locking system is implemented in the EIC RX as depicted by Fig. 1(b). It utilizes two main loops for wavelength adjustment: a DC loop adjusts the TIA output common-mode level by a DC current DAC (IDAC); a thermal loop tunes the CRR filter wavelength by a heater current DAC. The DC loop cancels the DC current from the photo diode (PD), which reflects the mismatch between the incoming wavelength and the filter's center wavelength. During the filter shape characterization, the heater loop sweeps through the heater DAC and concurrently reads the DC code. This process enables the system to classify the filter's shape and adopt the correct locking strategy.

The CRR filter characterization algorithm first searches for two DC peaks within its heater code sweeping range. If two DC peaks are found and the DC code at their center is lower than the DC peaks by a predefined threshold, this filter will be classified as a concave filter. The tracking algorithm will lock the filter to its local minimum, corresponding to the center of the peaks, as illustrated in case (i) of Fig. 1(b). The DC loop continuously monitors and adjusts the TIA DC level. Heater calibration is triggered when the instantaneous DC code exceeds the initial DC code by the calibration threshold. When calibration begins, the algorithm searches for and re-lock onto the local minimum.

In the alternative case, if two peaks are detected with a sufficiently shallow middle region, the filter is classified as a flat-top filter [case (ii) of Fig. 1(b)]. With only one peak identified, the algorithm measures the filter's 3-dB bandwidth. If its bandwidth exceeds the flat threshold, it is also categorized as a flat-top filter; otherwise, it is considered as a convex filter [case (iii) of Fig. 1(b)]. For both flat and convex filters, calibration is initiated when the instantaneous DC code drops by the calibration threshold. During calibration, the algorithm locates the DC peak by adjusting the heater code. For convex filters, it directly locks the DC peak, whereas for flat-top filters, it further moves the heater code by half of the 3-dB bandwidth from the DC peak position to center it appropriately.

The TX MRM wavelength locking process is analogous to the CRR locking mechanism. A DC loop continuously monitors and tunes the TX drop-port voltage by adjusting the drop-port current (I_{dp}). The TX thermal loop first sweeps through the heater code to find the peak I_{dp} . Once identified, the thermal loop continuously adapts the heater code to maintain around 75% of the peak current, optimizing the modulation efficiency and ensuring consistent performance.

4. Experiment and Results

Fig. 2 shows the measurement setup and chip photos. We first test RX CRR wavelength tracking by directing 0-dBm laser light from a tunable laser into the RX and sweeping its wavelength from 1308 nm to 1312 nm at 0.05 nm/min. With RX heater code fixed at 760, the PD current (I_{PD}) measured by the DC loop reveals the concave CRR spectrum centered at 1310 nm [Fig. 3(a)]. When the RX heater is activated, it automatically picks the concave-locking algorithm and adapts the heater code to stabilize I_{PD} at around 250 μ A, indicating consistent CRR alignment with its input wavelength. We then reduce the laser power to -3 dBm and repeat this measurement for a micro-ring resonator filter with a 1312-nm center wavelength, using a heater code of 756. The algorithm selects the peak-locking strategy, stabilizing I_{PD} at around 150 μ A [Fig. 3(b)]. Fig. 3(c) shows the heater code changes with wavelength variations.

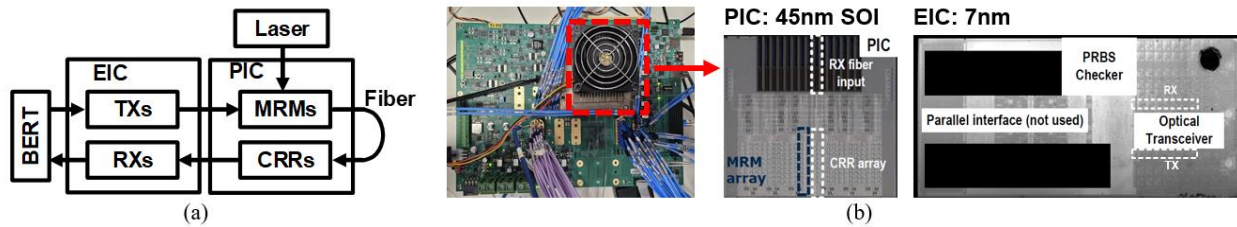


Fig. 2. (a) Experimental setup; (b) chip photos.

We then test the TX MRM tracking by directing the laser power into the TX that is modulated at 45 Gbps in a PRBS7 pattern, and measuring the TX eye with an oscilloscope. With a fixed heater code of 1598, the MRM optical modulation amplitude (OMA) reaches a maximum of 0.47 mW at 1310 nm, and quickly drops as the input wavelength deviates [Fig. 3(d)]. In contrast, with the activated TX heater, the OMA remains stable at around 0.47 mW through the entire sweeping range, its eye diagram maintains the shape in Fig. 3(e), and heater code tracks the wavelength shift shown by Fig. 3(c).

The TX-to-RX loopback with the CRRs is initialized at 1310 nm and 45 Gbps, achieving a BER below $1E-10$ which has reached the low limit of our BER measurement setup [Fig. 3(f)]. With both TX and RX heaters activated, the BER stays below $1E-10$ when the laser wavelength is swept from 1310 nm to 1312 nm at a rate of 0.025 nm/min. However, when either the TX or RX heater is turned off, the BER increases drastically even with 0.1-nm wavelength deviation. Given that the ring temperature sensitivity is around 0.0575 nm/ $^{\circ}$ C, this 0.1-nm wavelength shift corresponds to a mere 1.7 $^{\circ}$ C temperature change.

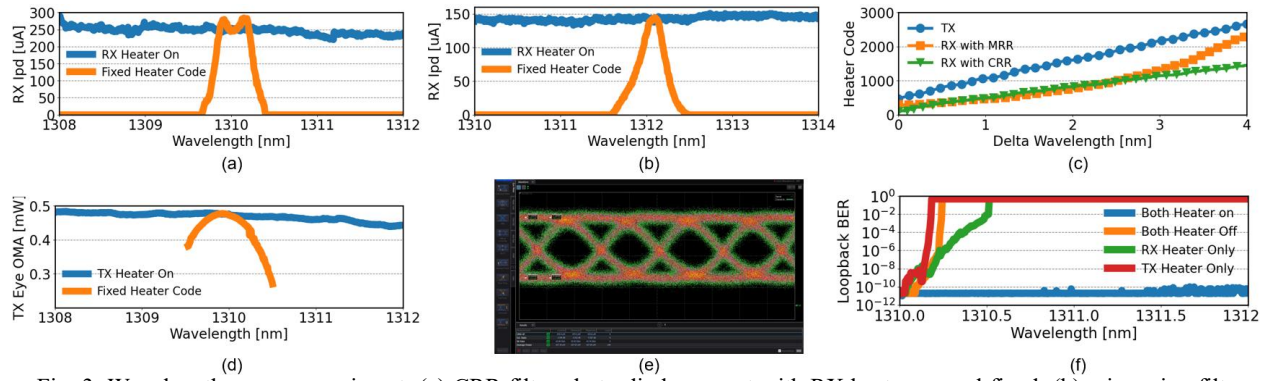


Fig. 3. Wavelength sweep experiment: (a) CRR filter photo-diode current with RX heater on and fixed; (b) micro-ring filter photo-diode current with RX heater on and fixed; (c) heater code versus delta wavelength; (d) TX drop-port voltage with TX heater on and fixed; (e) measured eye diagram with TX heater on; (f) BER versus wavelength under different heater setup.

To test the temperature tuning, we enable both the TX and RX heaters, and fix the laser wavelength at 1310 nm. We activate the heating circuits on the PIC to change its temperature from 38°C to 58°C, and back to 38°C, monitored by a temperature sensor on the EIC. Both the PD current and TX drop-port voltage are stable [Fig. 4(a)], and their heater codes are adjusted to counter the temperature shift [Fig. 4(b)]. As a result, the BER stays below 1E-10 across the entire temperature range while operating at 45 Gbps [Fig. 4(c)]. Fig. 4(d), (e), (f) are plots of the temperature change, heater code, and BER over time, respectively. The EIC power consumption is 54 mW, and the average power consumption of TX and RX heaters is 13.5mW, yielding an overall power efficiency of 1.5 pJ/bit.

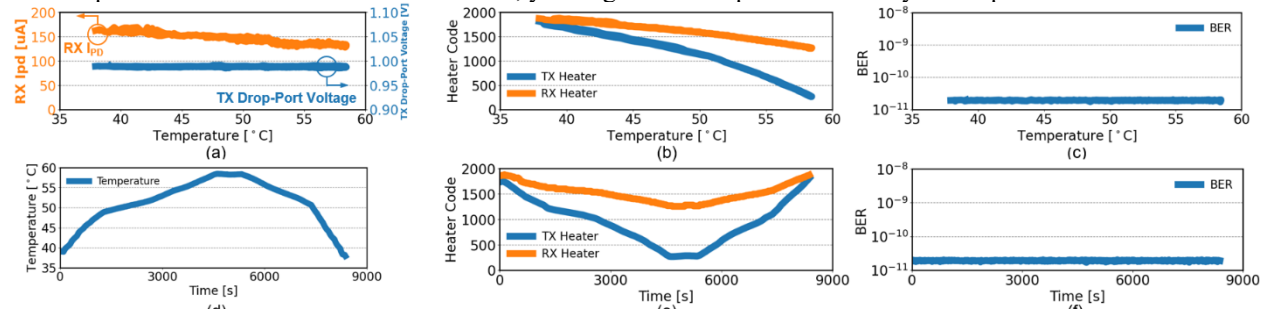


Fig. 4. (a) RX photo-diode current and TX drop-port voltage versus temperature; (b) TX and RX heater code versus temperature; (c) BER versus temperature; (d) temperature over time; (e) heater code over time; (f) BER over time.

5. Conclusion

This work demonstrates an automated and adaptive wavelength tracking and temperature stabilization system for the TX MRMs and RX CRR filters in a silicon photonics based DWDM link for co-packaged optics. The RX heater algorithm identifies and adapts to the complex profiles of CRR filters based on the DC loop data. The TX heater employs a similar approach to optimize its modulation efficiency. With the combined functionality of both heaters, the system reliably operates across a broad range of input wavelengths and temperature variations.

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6. References

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